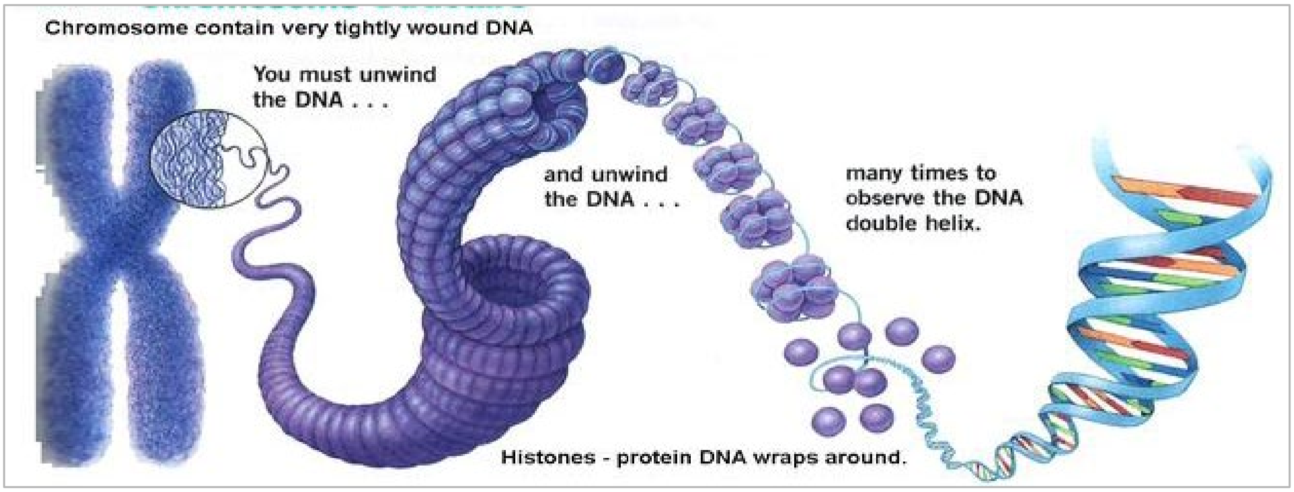
Name: ____________________________

UNIT 3A: CELL DIVISION

Topic 1: Mitosis

Chromosomes & Cell Division

DNA in the nucleus of each cell is tightly packaged into structures called chromosomes:



➢ *Explain the difference between chromosomes and chromatin. When would each be visible within a given cell?*

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Chromosome Number

- Chromosome number varies from species to species.
- In humans, somatic cells contain ___________ chromosomes (they are diploid, or _______), while gametes contain ___________ chromosomes (they are haploid, or _______).
- Species which contain more than two copies of each chromosome are __________________.

Name: ___________________________

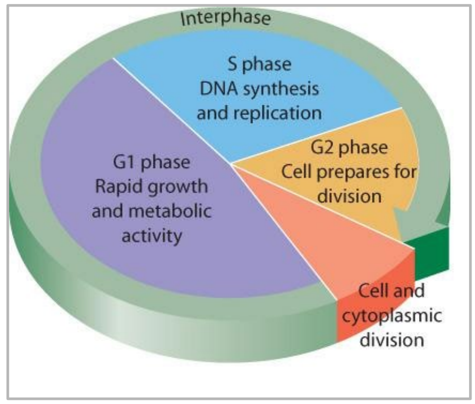
Define each of the following terms in the context of the human genetic material:

- Somatic cells
- Gametes
- Homologous chromosomes
- Autosomes
- Sex chromosomes
- Alleles
- Diploid
- Haploid

Name: ____________________________

The Cell Cycle

The goal of the cell cycle is to produce daughter cells with the same DNA as the parent cell. Each cell goes through three distinct phases before it actually divides:



1)

2)

3)

Collectively, these three phases are referred to as ________________________________.
Interphase is followed by ____, or
"____________________________".
M-phase accounts for the actual division of genetic information (______________________
and splitting of the cytoplasm itself (___________________________________).

Mitosis

Mitosis is performed by all somatic cells. It is a form of cell division that produces daughter cells that are identical to parent cells. List and explain the three general purposes of mitosis:

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-
-

Name: ___________________________

After DNA replication (interphase), cells must go through four stages of mitosis before cytokinesis can occur…

1. PROPHASE

2. METAPHASE

3. ANAPHASE

4. TELOPHASE

Name: ___________________________

➢ *Explain the difference between homologous chromosomes and sister chromatids. Use a diagram to illustrate your answer.*

➢ *How does mitosis in animal cells differ from mitosis in plant cells?*

Mitosis in Animal Cells	Mitosis in Plant Cells

Applications of the Cell Cycle

Thanks to an in-depth understanding of the cell cycle, we have been able to develop a number of useful technologies, including:

1)

2)

3)

Name: ______________________________

4)

1) CANCER TREATMENTS

- *Explain how you know if a sample of cells is cancerous.*

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- *Distinguish between mutagens and carcinogens.*

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2) CLONING

- *Explain the difference between monozygotic and dizygotic twins.*

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- *Use a flowchart to explain the process involved in artificially cloning an organism:*

Name: ____________________________

3) STEM CELL RESEARCH

- *What is a stem cell? Outline some of the uses of stem cell research.*

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- *Distinguish between totipotent and pluripotent stem cells.*

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4) TREATMENTS FOR AGING?

- *How can telomeres be used to determine a cell's age?*

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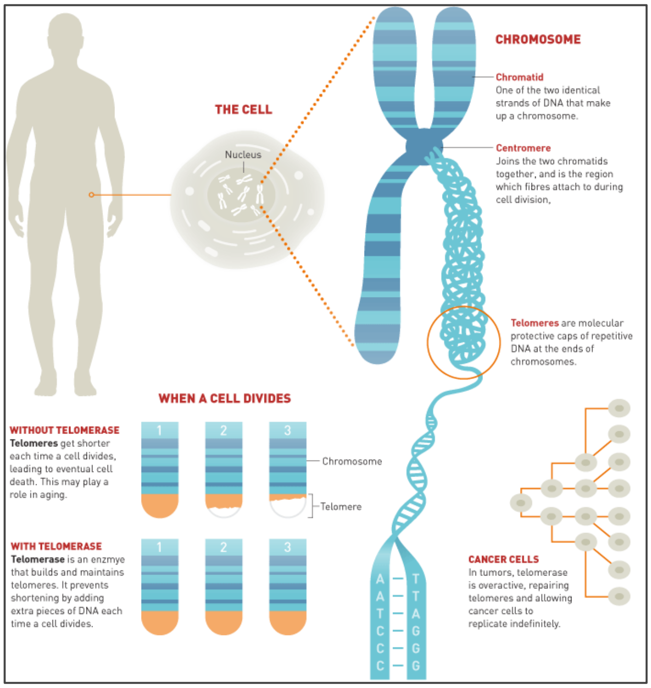
Name: ______________________________



THE CELL
Nucleus
CHROMOSOME
Chromatid
One of the two identical strands of DNA that make up a chromosome.
Centromere
Joins the two chromatids together, and is the region which fibres attach to during cell division,
Telomeres are molecular protective caps of repetitive DNA at the ends of chromosomes.
WHEN A CELL DIVIDES
WITHOUT TELOMERASE
Telomeres get shorter each time a cell divides, leading to eventual cell death. This may play a role in aging.
1
2
3
Chromosome
Telomere
WITH TELOMERASE
Telomerase is an enzmye that builds and maintains telomeres. It prevents shortening by adding extra pieces of DNA each time a cell divides.
1
2
3
A A T C C C
T T A G G G
CANCER CELLS
In tumors, telomerase is overactive, repairing telomeres and allowing cancer cells to replicate indefinitely.